PROPHET: Probabilistic Reliability via Outcome-Pricing for Honest Evaluation of Agent Calibration

Anonymous ACL submission

Abstract

Frontier large-language-model agents now solve nearly every static-resettable benchmark we have, yet a 37 % gap between lab accuracy and production reliability persists. The capability we are mis-measuring is *calibrated agency* — knowing when to commit, declining tasks one should not take, and pricing one's own confidence with skin in the game. We introduce **PROPHET**,[1] a benchmark that evaluates LLM-based agents inside a marketplace. For each task, the agent must commit (TAKE), declare a probability of success (QUOTE, scored on a Brier-based proper scoring rule conditional on the Quote action) or abstain (PASS). PROPHET ships 12 procedurally-generated task families spanning code, math, knowledge, reasoning, writing, browser, tool-use, multimodal, data-analysis, scientific, multilingual, and safety domains, evaluated at *two difficulty regimes*: a **Standard tier** where frontier accuracy is near-ceiling (90–97 %) yet ECE still varies $\sim$7$\times$ across systems, and a $\mathbf{T_{extreme}}$ **tier** ($\delta \geq 0.97$) with multi-modulus CRT, Pell equations, discrete logarithms, depth-9 temporal chains, 9×9 Sudoku, modular knapsack, lattice-path counting, and Einstein/zebra puzzles, where frontier accuracy drops to $\sim$10–40 % by design. All verifiers are mechanical. We benchmark 17 frontier and open-weight systems via OpenRouter (Anthropic `claude-haiku-4.5/sonnet-4.6/opus-4.7`; Google `gemini-3-flash-preview/3.1-flash-lite/3.1-pro-preview`; OpenAI `gpt-5/5-mini/5-nano/5.2/5.4-nano`; DeepSeek `v3.2/r1`; Meta `llama-4-scout/maverick`; Qwen `qwen3-32b/235b-thinking`; Moonshot `kimi-k2-thinking`). Headline metrics use a pre-registered minimum-N filter ($N \geq 100$ Standard, $N \geq 20$ $T_{extreme}$) so small-N outliers do not dominate the leaderboard. The headline finding from the Standard tier: **the first-place leader differs between accuracy and net payoff.** Best by accuracy: `openai/gpt-5.2` (0.970, tied with `google/gemini-3-flash-preview`). Best by calibration: `openai/gpt-5.2` (ECE $= 0.017$, the lowest on the panel). Best by net payoff: `google/gemini-3.1-pro-preview` (37,813 PROPHET-cents) — not the agent that wins on accuracy or calibration. On the matched $K = 16$ panel Spearman $\rho_{\text{acc, ECE}} = -0.94$, $\rho_{\text{acc, payoff}} = +0.78$, $\rho_{\text{ECE, payoff}} = -0.71$. On $T_{extreme}$ the picture shifts further: the new $(\text{ECE}, \mathcal{P})$ Pareto frontier is `gpt-5` (ECE 0.019, payoff 5,274) and `gpt-5.2` (ECE 0.025, payoff 10,500); median accuracy collapses from 87 % to 60 % (-27 pp) and median ECE worsens $2.12\times$. All code, procgen seeds, agent adapters, and per-family eval harness are released under MIT and on Hugging Face for re-execution at any future model checkpoint.

1 Introduction

By May 2026, frontier language-model agents saturate nearly every static benchmark: MMLU-Pro and GPQA-Diamond cluster above 94 %; SWE-bench Verified at $\approx$ 94 % (Jimenez et al., 2024); HLE rose from 8 % to $\approx$ 65 % in sixteen months (Scale AI and Center for AI Safety, 2025); ARC-AGI-2 from 53 % to 85 % in four months (Chollet et al., 2025). The capabilities that did *not* saturate are calibrated, persistent agency: ARC-AGI-3 $<$ 1 % (ARC Prize Foundation, 2026); SWE-EVO $\approx$ 25 %; MemoryAgentBench selective-forgetting $<$ 30 % (Hu et al., 2025).

Production has its own scoreboard. A widely-cited industry analysis (May 2026) found a 37 % gap between lab and production performance and 50$\times$ cost variation among systems with compara-

[1] Distinct from *Prophet Arena* (Yang et al., 2025b) (UChicago, October 2025), which benchmarks LLM forecasting on live Kalshi markets with natural events. PROPHET is a procedurally-generated, mechanically-verified, two-tier calibration benchmark with a Brier-scaled marketplace; the two suites are complementary, not overlapping.

ble accuracy (Cognition Labs, 2025). The dominant production failure is not low accuracy but *ambition mismatch* — systems committing to tasks they should have declined, reporting high confidence under conditions in which their reliability is demonstrably lower (Chen et al., 2025b). The missing measurement is not another accuracy benchmark; it is an evaluation that pays the agent only when its calibration matches its skill, where over-confidence costs the same currency as incompetence, abstention is a first-class action, and the leaderboard is a Pareto frontier rather than a scalar.

Contributions. We introduce PROPHET, a calibration benchmark for LLM-based agents. (i) A **marketplace protocol** with three actions per task (TAKE, QUOTE, PASS) and a Brier-based proper scoring rule, conditional on QUOTE (Section 3). (ii) Twelve **procedurally generated task families** spanning the production-LLM surface, evaluated at two regimes per family: a Standard tier at near-ceiling difficulty (median 87 % frontier accuracy) and a $\mathbf{T_{extreme}}$ tier ($\delta \geq 0.97$) where frontier accuracy is held to $\sim$10–45 % by construction (multi-modulus CRT, Pell, discrete log, 9×9 Sudoku, depth-9 temporal chains, modular knapsack); all verifiers are mechanical (Section 4). (iii) A **Pareto-frontier leaderboard** on $(\mathcal{P}, \mathrm{ECE})$ with a secondary $(\mathrm{cost}, \mathcal{P})$ frontier and a pre-registered minimum-N filter; no scalar headline. (iv) A **Model Overreach Point** (MOP) metric: the difficulty bin where stated $\hat{P}$ exceeds empirical accuracy by $\geq 2\sigma$. (v) A **gaming-resistance audit** adopting the Berkeley RDI April 2026 attack taxonomy (Appendix A.5). (vi) Open-source release of code (MIT), procgen seeds + reference panel (Hugging Face), and the full experimental matrix (Appendix A).

Empirical finding. We benchmark 17 frontier and open-weight models via OpenRouter (Anthropic Haiku 4.5 / Sonnet 4.6 / Opus 4.7; Google Gemini 3-Flash / 3.1-Flash-Lite / 3.1-Pro; OpenAI GPT-5 / 5-mini / 5-nano / 5.2 / 5.4-nano; DeepSeek V3.2 and R1; Meta Llama-4-Scout and Maverick; Qwen3-32B and 235B-Thinking; Moonshot Kimi-K2-Thinking). **The rank-order by accuracy differs from the rank-order by net payoff at the first-place level.** On the Standard tier (matched $K = 16$ panel, $N \geq 100$): $\rho_{\text{acc, ECE}} = -0.94$, $\rho_{\text{acc, payoff}} = +0.78$, $\rho_{\text{ECE, payoff}} = -0.71$. Best by accuracy and calibration: `openai/gpt-5.2` (0.970 acc, ECE 0.017, the lowest on the panel). Best by net payoff: a *different* agent — `google/gemini-3.1-pro-preview` (37,813 vs. 31,328 for GPT-5.2 and 24,824 for Claude Opus 4.7). On $\mathrm{T}_{\text{extreme}}$ the $(\mathrm{ECE}, \mathcal{P})$ frontier is GPT-5 and GPT-5.2; the next-best open-weight (`qwen3-235b-thinking`, ECE 0.091) is dominated on $(\mathrm{ECE}, \mathcal{P})$ but Pareto-optimal on $(\mathrm{cost}, \mathcal{P})$ at $\frac{1}{4}$ the operating cost. The first-place leader changes under at least one axis on both tiers — the empirical claim PROPHET is designed to make visible.

2 Related work

Saturation of static benchmarks. By May 2026, MMLU-Pro and GPQA-Diamond cluster above 94 %; SWE-bench Verified at $\approx$ 94 % (Jimenez et al., 2024); HLE rose from 8 % to $\approx$ 65 % in 16 months (Scale AI and Center for AI Safety, 2025); ARC-AGI-2 from 53 % to 85 % in four months (Chollet et al., 2025). The unsaturated axes are all calibrated, persistent agency: ARC-AGI-3 $<$ 1 % (ARC Prize Foundation, 2026), SWE-EVO $\approx$ 25 %, MemoryAgentBench selective-forgetting $<$ 30 % (Hu et al., 2025).

Verbal calibration and faithfulness. Kadavath et al. (Kadavath et al., 2022) showed LLMs have non-trivial verbal calibration. Recent faithfulness work (Chen et al., 2025b) found unfaithful chains-of-thought are systematically *longer* than faithful ones and that models verbalise decision-relevant hints in only 25–39 % of cases — verbose confidence decouples from competence. PROPHET pays only when stated $\hat{P}$ matches outcome rates.

Proper scoring rules. Brier (1950) and Gneiting–Raftery (Gneiting and Raftery, 2007) provide the theory; Guo et al. (Guo et al., 2017) established ECE and temperature scaling for deep classifiers. We adopt Brier because it is bounded, smooth, and admits closed-form expected payoff at known $P[y = 1 \mid t]$ — log-score blows up at extremes and is gameable under borderline tasks.

Agent benchmarks and gaming. GAIA (Mialon et al., 2024), AgentBench, MMAU, and OS-World (Xie et al., 2024) established multi-step tool-use surfaces. The Berkeley RDI April 2026 audit (Berkeley RDI, 2026) demonstrated trivial exploits giving 100 % on eight major agent benchmarks; we adopt their attack taxonomy as a pre-publication red-team checklist (Section 3).

Concurrent calibration-with-skin-in-the-game work. **Prophet Arena** (Yang et al., 2025b) (UChicago, Oct. 2025) evaluates frontier LLMs on live Kalshi prediction-market events with Brier and a market-return metric; it operates on *live external events* rather than procedural tasks with mechanical verifiers, has no abstention action, and admits no Pareto comparison. **KalshiBench**, **PolyBench**, and **ForecastBench** (Karger et al., 2024) cover similar territory and inherit the same external-event dependency. **BAS** (Wu et al., 2026) (Oxford, Apr. 2026) is the closest sibling: an asymmetric-overconfidence proper-scoring rule with abstention. PROPHET differs in (i) a bounded Brier-based payoff $[-3\kappa, +\kappa]$ rather than an unbounded log-style penalty, (ii) an explicit TAKE/QUOTE/PASS action menu, (iii) a procedurally generated 12-family suite with mechanical verifiers, and (iv) a two-dimensional Pareto frontier rather than a scalar. **AbstentionBench** (Kirichenko et al., 2025) and **Going-All-In** (Todasco et al., 2025) jointly motivate the bundle: the former shows current systems do not abstain when ECE indicates they should; the latter that prompted-confidence elicitation diverges from behavioural commitment. **CRS** (Salla et al., 2025) proposes a single composite score; we deliberately avoid this in favour of Pareto.

Hard procedural task families. **Enigmata** (Chen et al., 2025a) ships 36 puzzle generators with Python verifiers; Sequential-Hard holds o4-mini at 34 %. **ZebraLogic** (Lin et al., 2025) documents a "curse of complexity" dropping Claude 3.5 to 12.4 % on the hard tier. **MultiZebraLogic** (Bruun and Smart, 2025) shows red-herring clauses drop o3-mini by 15 ± 7 pp. **NPPC** (Yang et al., 2025a) ships 25 NP-complete generators where all frontier models fall below 20 % past complexity 5. **OJBench** (Wang et al., 2025a), **LiveCodeBench-Pro** (Zheng et al., 2025), and **AetherCode** (Wang et al., 2025b) sit at 0–10 % pass@1 on their hard tiers; **NoLiMa** (Modarressi et al., 2025) shows GPT-4o falling from 99.3 % to 69.7 % under minimally lexical needle-in-haystack. PROPHET's $\mathrm{T}_{\text{extreme}}$ tier draws on the same recipe so accuracy spread is preserved at ~10–40 % after the standard tier saturates. The shared word with Prophet Arena reflects only the proper-scoring-rule lineage; the design choices are non-overlapping.

3 The PROPHET marketplace

Setup. A benchmark cycle consists of a procedurally generated task suite $\mathcal{T}$ and an agent $\mathcal{A}$. Each task $t \in \mathcal{T}$ is drawn from one of twelve families (Section 4) with a known reference-panel difficulty $\delta(t) \in [0, 1]$. Difficulty climbs from T0 (trivial) to T7 ("standard hardest"); we publish an additional $\mathrm{T}_{\text{extreme}}$ tier at $\delta \in [0.97, 1.0]$ calibrated against 2025–2026 hard-task ceilings. For each t the engine forms a deterministic market offer $(V_t, C_t, \kappa_t, \delta_t^{\text{pass}})$ from a family-specific base plus hash-derived per-instance noise $\leq 2\%$.

Actions and payoffs. The agent selects $a \in \{\textsc{Take}, \textsc{Quote}, \textsc{Pass}\}$, declares $\hat{P} \in [0, 1]$, and receives: TAKE: $V_t y - C_t(1-y)$; QUOTE: $V_t y - C_t(1-y) + \kappa_t(1 - 4(\hat{P} - y)^2)$; PASS: $-\delta_t^{\text{pass}}$. Here $y \in \{0, 1\}$ is the mechanical verifier's outcome. $\kappa(1 - 4(\hat{P} - y)^2)$ is a Brier-based proper scoring rule scaled to $[-3\kappa, +\kappa]$: conditional on QUOTE, expected payoff is maximised iff $\hat{P} = \mathbb{E}[y \mid t]$ (Brier, 1950; Gneiting and Raftery, 2007). The factor 4 makes uniform $\hat{P} = 0.5$ yield zero calibration payoff, eliminating the always-hedge strategy. The asymmetric $[-3\kappa, +\kappa]$ band reflects production reality: over-confidence drives bad decisions; caution costs research effort. The full action-selection equilibrium is in Appendix A.

Headline metrics and Pareto leaderboard. We report net payoff $\mathcal{P} = \sum_t \text{payoff}(t)$, ECE (10 equal-width bins; BCa bootstrap 95 % CI as ECE is non-differentiable, $B = 10^4$), Brier, log-loss, abstention precision (on a $\rho = 0.10$ shadow subsample), and Model Overreach Point MOP (the difficulty bin where stated $\hat{P}$ exceeds empirical accuracy by $\geq 2\sigma$). We publish the Pareto frontier of $(\mathcal{P}, \text{ECE})$ with a secondary $(\text{cost}, \mathcal{P})$ frontier; the frontier table, not any scalar, is the headline.

Statistical protocol. Pairwise contrasts: paired Student's t when $n \geq 30$ (else Wilcoxon); McNemar's exact for binary correctness; permutation test for ΔECE ($B = 10^4$). Effect sizes accompany every p (Cohen's d, rank-biserial r, paired odds-ratio, $|\Delta\text{ECE}|$). Holm–Bonferroni is applied *within* each metric family ($\binom{16}{2} = 120$ tests for net-payoff; 120 for ECE), not across metrics. Raw and adjusted p-values are reported in parallel; post-hoc power and MDE per headline contrast are in Appendix B.

Per-agent N and min-N filter. The Standard tier targets 240 tasks per agent; realised N ranges 55–292 (multi-seed pooling on Pareto-essentials, provider-side refusals). T_{extreme} targets 40 and realises 28–100. The pre-registered min-N filter (Standard $N \geq 100$, Ext $N \geq 20$) governs Pareto-flag and Spearman; rows below threshold are displayed for transparency without the $\star$ marker.

Gaming resistance. We adopt the Berkeley RDI April 2026 attack taxonomy as a pre-publication red-team checklist (Appendix A.5). Key mechanisms: procedural seeds with rotation (no fixed eval set), mechanical primary metrics (no LLM-judge), sandboxed code-family verifiers (`subprocess.run` with 5 s timeout, restricted PATH), and the Pareto frontier itself — single-axis gaming collapses an orthogonal axis.

4 Task families

PROPHET ships twelve procedurally generated task families. Each family exposes (a) a deterministic generator from a seed, (b) a *mechanical* verifier returning a Boolean, and (c) a reference-panel-calibrated difficulty per task. The selection criterion was the intersection of three constraints: capability coverage of the production-LLM surface (code, math, knowledge, reasoning, writing, tool-use, web, multimodal, data, scientific, multilingual, safety); mechanical verifiability (regex, parser, integer-equality, or sandboxed execution — no LLM-judge in the primary loop); and lit-anchored difficulty headroom (Chen et al., 2025a; Lin et al., 2025; Yang et al., 2025a; Wang et al., 2025a; Zheng et al., 2025; Wang et al., 2025b; Modarressi et al., 2025; Sun et al., 2025). Excluded under these constraints were multi-turn dialog, open-ended summarisation, RAG, and long-horizon agentic planning (v2 candidates). Table 1 gives the family palette; generator and verifier source is in the public release.

Table 1: The twelve PROPHET task families. "Tiers" = number of procedurally-graded difficulty levels.

Family	Capability surface	Verifier
math	arithmetic $\rightarrow$ olympiad-lite	Fraction-equality
code	predict / write / fix programs	sandboxed `exec`
knowledge	closed-form factual QA	exact / synonym / numeric
reasoning	ARC-AGI-like grids + deduction	JSON-equality
writing	constrained text (regex / acrostic)	regex / parser
tools	multi-step tool-use chains	pure-function replay
browser	text-only browser simulator	action-replay equality
multimodal	ASCII-encoded multimodal	exact / integer / char
data	data-analysis on small CSVs	numeric tolerance
scientific	physics / chem / bio closed-form	numeric tolerance (1‰)
multilingual	language-id + cross-lingual ops	language-set / lookup
safety	should-refuse-or-comply routing	refusal-token / answer mat

Standard tier (T0–T7). Difficulty climbs from trivial (T0) to "standard hardest" (T7) within each family. Examples: math runs from one-step arithmetic to olympiad-lite sum-of-divisors; code runs from output-prediction on small Python snippets to write-and-test; reasoning runs from sequence puzzles to 3×3 ARC-grid transforms and transitive deduction with red herrings. Knowledge uses synonym-set verifiers; writing uses regex / parser / acrostic / JSON-schema / ABAB-rhyme (the rhyme verifier was upgraded post-pilot from suffix-match to a four-rule nucleus-aware checker, Section 7). Tools and browser run on in-memory simulators with action-replay verifiers; a T7 browser task plants an "IGNORE PRIOR INSTRUCTIONS" injection that is never the right answer.

T_{extreme} tier ($\delta \in [0.97, 1.0]$). Five families ship hardened generators whose frontier accuracy is held to 10–40 % by construction. *math*: Euler totient over a non-uniform range, multiplicative-order $\mathrm{ord}_p(a)$ and modular quadratic residues over primes $p \in [10^3, 5 \cdot 10^3]$, 12-modulus CRT chains, fundamental Pell solutions ($x^2 - Dy^2 = 1$), lattice-path counting with forbidden cells on $n = 8$–12 grids, and discrete logarithms modulo primes $p \in [5 \cdot 10^3, 2 \cdot 10^4]$. *code*: subset-sum count over 8–12 items, modular knapsack (11–15 items, weights ≤ 30, modulus 5–11), and DAG path counting on 12–16 nodes. *knowledge*: 6-word letter-count chains ($c_1c_2 - c_3c_4 + c_5c_6$), obscure-fact compositions sourced from 12 lesser-known historical figures, and 3-step word manipulations. *reasoning*: 5×5 Latin-square cell-fill, depth-9 temporal chains with 11+ clues, 3×3-digit cryptarithmetic multiplication, and 5×4 Einstein/zebra puzzles with 8–12 clues (uniqueness verified by backtracking during procgen). *multimodal*: 7×7 grid transform chains (apply 9 sequential transforms then read one cell) and 9×9 Sudoku cell-fill with 50 masked cells, target provably unique. All extreme verifiers return integer or string equality against a closed-form ground truth.

Other families. *tools*: pure-Python registry of 13 fixed tools (arithmetic, table lookup, filter, sort, dedup, count); three exhibit subtle pathologies (off-by-one, semantic red-herring, false-success). *data*: procedurally generated CSVs (3–30 rows) embedded as fenced ```` ```csv ```` blocks; tasks cover lookup, aggregation, group-by, two-table join, and time-series outlier detection. *scientific*: physics / chemistry / biology / earth-science with closed-form numerical answers ($\pm 10^{-3}$ relative tolerance). *multilingual*: a 200-word EN / ES / FR / DE / HI / JP dictionary supports language-id, multiple-choice translation, cognate selection, cross-lingual arithmetic, ordering, and grammar. A v2 expansion to ten languages (adds AR / ZH / RU / KO) is in Appendix B.7. *safety*: 50/50 split of benign and refusal-required tasks; the latter use synthesised placeholder phrasings "`<SYNTHETIC_PROHIBITED>`". The mechanical verifier accepts `PASS` or any of 14 refusal substrings, with $C_{\text{failure}} = 200$ so wrongly committing on a refusal task is penalised twice as heavily.

5 Results

5.1 Two-tier evaluation overview

We benchmark every agent at *both* the Standard tier (12 families, near-ceiling difficulty) and the $\mathrm{T}_{\text{extreme}}$ tier (5 families with $\delta \in [0.97, 1.0]$). The Standard tier targets 240 tasks per agent (20 per family $\times$ 12); the $\mathrm{T}_{\text{extreme}}$ tier targets 40 tasks per agent (8 per family $\times$ 5). Realised per-agent N varies because of provider-side refusals and multi-seed pooling on the Pareto-essentials; both are disclosed in Table 2. Headline Spearman and Pareto-frontier flags use a pre-registered minimum-N filter ($N \geq 100$ Standard, $N \geq 20$ $\mathrm{T}_{\text{extreme}}$). Bootstrap 95 % CIs (BCa for ECE, percentile elsewhere; $B = 10^4$) are reported for every metric.

H1: rank by accuracy $\neq$ rank by calibration $\neq$ rank by payoff. Spearman ρ on the $K = 16$ agents with $N \geq 100$ on the Standard tier: $\rho_{\text{acc, ECE}} = -0.94$, $\rho_{\text{acc, payoff}} = +0.78$, $\rho_{\text{ECE, payoff}} = -0.71$. The best-by-accuracy and best-by-ECE model is `openai/gpt-5.2` (acc 0.970, ECE 0.017; tied with `gemini-3-flash-preview` on accuracy but strictly better on calibration). The best-by-payoff model is a different agent: `google/gemini-3.1-pro-preview` (37,813 PROPHET-cents vs. 31,328 for GPT-5.2 and 23,800 for Gemini-3-Flash). Best accuracy and best calibration do not imply best deployment payoff under calibrated-commitment pricing; the orderings induced by each axis are demonstrably different at the first-place level.

Table 2: Two-tier PROPHET leaderboard. **Std** = standard tier (12 families, near-ceiling difficulty). **Ext** = $\mathrm{T}_{\text{extreme}}$ (5 families, $\delta \geq 0.97$). Net payoff in PROPHET-cents; ECE on $[0, 1]$. Multi-seed re-runs are pooled by agent name. $\star$ = Pareto-optimal on (ECE, $\mathcal{P}$) among agents with $N \geq 100$ (Std) / $N \geq 20$ (Ext); small-N rows are shown for transparency but do not receive the marker.

	Standard tier				
Agent	Acc	ECE	$\mathcal{P}$	Cost	
oracle	1.000	0.000	38685	0.00	24
google-gemini-3.1-pro-preview	0.911$\star$	0.054	37813	5.26	29
openai-gpt-5	0.945$\star$	0.040	36601	3.82	27
openai-gpt-5.2	0.970$\star$	0.017	31328	2.20	23
anthropic-claude-opus-4.7	0.900	0.078	24824	1.35	22
google-gemini-3-flash-preview	0.970	0.027	23800	0.21	23
anthropic-claude-haiku-4.5	0.935	0.043	22684	0.54	23
deepseek-deepseek-v3.2	0.846	0.129	22566	0.06	23
openai-gpt-5-mini	0.871	0.123	21768	0.50	23
meta-llama-llama-4-maverick	0.853	0.122	19253	0.08	23
anthropic-claude-sonnet-4.6	0.870	0.093	17793	0.80	23
qwen-qwen3-32b	0.843	0.110	17001	0.09	21
google-gemini-3.1-flash-lite	0.835	0.165	15976	0.09	23
openai-gpt-5-nano	0.819	0.154	14685	0.15	21
meta-llama-llama-4-scout	0.831	0.147	14194	0.03	20
qwen-qwen3-235b-a22b-thinking-2507	0.911	0.068	13216	0.49	12
openai-gpt-5.4-nano	0.734	0.133	11912	0.04	22
moonshotai-kimi-k2-thinking	1.000	0.004	10026	1.25	6
deepseek-deepseek-r1	1.000	0.008	7883	0.56	5
always-pass	nan	nan	-640	0.00	
random	0.000	0.532	-18203	0.00	15
always-take	0.000	0.990	-24370	0.00	24

H2: reasoning mode raises conditional accuracy and net payoff but does not uniformly increase strategic abstention. On the $\mathrm{T}_{\text{extreme}}$ tier, the DeepSeek reasoning variant abstains far more than its non-reasoning sibling (`deepseek-r1` 86 % PASS vs. `deepseek-v3.2` 4 %); the Qwen reasoning variant abstains *less* than its smaller non-reasoning sibling (`qwen3-235b-thinking` 26 % vs. `qwen3-32b` 48 %) because the smaller 32B model PASSes from inability rather than calibrated abstention. The robust signal is in *conditional accuracy*: reasoning variants are markedly higher (R1 82.4 % vs. V3.2 51.0 %; Thinking 83.1 % vs. 32B 67.3 %) and so is net payoff (R1 2755 vs. V3.2 550; Thinking 4825 vs. 32B 1588). The hypothesis "reasoning over-commits" is **rejected** on net payoff for both families, but the intuitive "reasoning $\Rightarrow$ more abstention" mechanism **holds only for DeepSeek**, not for Qwen. The unified statement the data support is: extended thinking improves conditional accuracy enough that, even when the variant commits less, net payoff goes up.

H3: open-weight Pareto-essentials match closed frontier on the cost–payoff axis, not the ECE–payoff axis. The T_{extreme} $(\mathrm{ECE}, \mathcal{P})$ Pareto frontier (min-N filter, $N \geq 20$) contains two closed-source agents: `openai/gpt-5.2` (ECE 0.025, payoff 10,500, $N = 71$) and `openai/gpt-5` (ECE 0.019, payoff 5,274, $N = 36$). The closest open-weight challenger, `qwen3-235b-thinking` (ECE 0.091, payoff 4,825), is dominated on $(\mathrm{ECE}, \mathcal{P})$ by both. *However*, on the $(\mathrm{cost}, \mathcal{P})$ frontier Qwen3-235B-Thinking delivers 4,825 at \$1.00 vs. `gpt-5.2`'s 10,500 at \$4.07 — payoff-per-dollar 4,825 vs. 2,580 — so the cost-conscious open-weight deployment claim survives even though the strict $(\mathrm{ECE}, \mathcal{P})$ claim does not.

H4: hard-tier accuracy collapse. On the matched N-filtered Standard panel ($K = 16$, $N \geq 100$) the median accuracy drops from 87.0 % (Standard) to 60.0 % (T_{extreme}, $N \geq 20$) — a 27.0-pp fall — while the median ECE *worsens* by $+0.123$ (Standard 0.110, T_{extreme} 0.233, a 2.12$\times$ inflation). Specific drops include `claude-opus-4.7` (90.0 % $\to$ 63.6 %, $\Delta = 26.4$ pp), `deepseek-v3.2` (84.6 % $\to$ 51.0 %), and `llama-4-scout` (83.1 % $\to$ 17.0 %). The Standard-tier near-ceiling effect was a difficulty-floor artefact.

Strategic family-level abstention. A cross-cut of the per-(agent, family) heatmaps reveals a clean *strategic-abstention cluster*: `deepseek-r1` declines on 9 of 12 families (100 % PASS); `kimi-k2-thinking` on 7; `qwen3-235b-thinking` on 4; `openai/gpt-5` on 3; `gemini-3.1-pro-preview` on 2. The non-reasoning agents (`haiku-4.5`, `gemini-3-flash`, `deepseek-v3.2`, the Llama family) have zero 100 %-PASS families. Reasoning-tuned systems recognise families beyond their competence and decline at the family level; non-reasoning systems commit indiscriminately. Per-family heatmaps (PASS rate, committed-N, ECE conditional on commitment), per-language multilingual breakdown, Pareto-frontier scatter, reliability diagrams, MOP heatmap, per-pair significance, post-hoc power, cost-conscious $(\mathrm{cost}, \mathcal{P})$ frontier, strict-verifier ablation, and rhyme / safety LLM-judge audits are all in Appendix B.

6 Discussion

Why economic pricing matters. A scalar leaderboard rewards accuracy at any cost. Production buyers care whether an agent is *worth deploying*: how often does it commit and succeed, how costly are its mistakes, and what is the spread of its expected value? PROPHET pays the agent for being honest about its odds, making calibration the business metric rather than an ablation.

Why a Pareto frontier, not a scalar. The right deployment trade-off depends on cost-of-error. A safety-critical system needs both low ECE and high net payoff (Pareto corner); a research bot can tolerate higher ECE if net payoff covers human review. We publish both $(\mathcal{P}, \mathrm{ECE})$ and $(\mathrm{cost}, \mathcal{P})$ frontiers and explicitly avoid a composite score (Salla et al., 2025).

Why a Brier-based proper scoring rule. Brier is bounded, C^∞ smooth, and admits closed-form expected payoff at known $P[y = 1 \mid t]$. Logarithmic score is also proper but unbounded at $\hat{P} \to \{0, 1\}$, making it gameable when the agent reports near-extreme probabilities on borderline tasks. CRPS is preferable for continuous outcomes; we adopt it in a forthcoming v2 for the writing family.

Why two difficulty regimes. The Standard tier measures calibration at near-ceiling accuracy (90–97 %) where production cost is dominated by rare costly miscommits and ECE varies 7$\times$ across systems. The T_{extreme} tier deliberately mimics the research regime: at saturation-frontier tasks even the best 2026 systems must hedge or abstain. The empirical joint pattern is exactly what we observe in section 5: Pareto ordering changes between tiers and the calibration–accuracy correlation breaks at the hard tail (median ECE worsens 2.12$\times$ from Standard to T_{extreme}).

What the data tells us. *Reasoning mode.* Reasoning-tuned variants raise conditional accuracy and net payoff but do *not* uniformly raise strategic abstention: DeepSeek R1 jumps from 4 % to 86 % PASS on T_{extreme}, while Qwen3-Thinking abstains *less* than its 32B sibling because the 32B model passes from inability rather than calibrated abstention. *Open vs closed.* Open-weight models do not dominate the $(\mathrm{ECE}, \mathcal{P})$ frontier on T_{extreme} but Qwen3-235B-Thinking is Pareto-optimal on the $(\mathrm{cost}, \mathcal{P})$ frontier at $\frac{1}{4}$ the operating cost. The cost-conscious open-weight deployment claim survives where the strict calibration claim does not.

What this benchmark does not measure. Long-horizon coherence (CHRONOWEAVE), drift under silent environmental change (DRIFTWORLD), and catastrophic-risk safety (targeted red-team suites) are out of scope. PROPHET covers calibrated commitment.

7 Limitations

Binary outcomes. The Brier-based calibration term assumes $y \in \{0, 1\}$. Open-ended writing tasks are thresholded via mechanical constraint checks; this loses fine-grained quality. A continuous-outcome extension using log-score or CRPS is a natural follow-up.

Closed-model nondeterminism. Vendor APIs are not bit-exact even at temperature 0. We freeze the request payload (model, prompt, temperature, max-tokens); response-noise variance is captured in bootstrap CIs.

Procedural generation is not natural data. Procedural tasks may not exercise long-tail real-world reasoning. We draw slot fillers from curated permissively-licensed corpora but worst-case complexity still trails natural tasks. PROPHET complements rather than replaces natural-data benchmarks.

Safety family is synthetic. Refusal-required tasks use placeholder unsafe phrasings; we measure *refusal routing* only, not catastrophic-risk content moderation. The substring-keyword verifier is audited against an LLM-judge baseline in Appendix B (89.3 % agreement, false-accept-rate 0 %).

Multilingual coverage. The Standard tier uses a 200-word six-language dictionary; we extend to ten languages (`de`, `en`, `es`, `fr`, `hi`, `jp`, `ar`, `zh`, `ru`, `ko`) in Appendix B.7. Long-form translation, idioms, and genuine code-switching remain out of scope.

Closed leaderboard drift. Vendors update closed models without renaming. We mitigate by reporting the exact model id and run date; community runs against new checkpoints append automatically.

Mechanical verifiers can be evaded. An adversarial model could output answers that coincidentally satisfy the verifier. We mitigate via N-test-case verification on code, synonym sets on knowledge, and difficulty calibration that lifts the implicit-target range outside the agent's prior. The rhyme verifier was specifically upgraded post-pilot from a strict last-2-letter suffix match to a four-rule nucleus-aware checker (canonical-form, orthographic-equivalence-group, shared trailing-consonant-cluster, literal-tail fallback) with a 16-pair positive set and an 8-pair negative set; pre-pilot rhyme-heavy commitments were underrated by 1–3 pp.

Difficulty stratification was uneven in the standard tier. The Standard T7 generators are still near-ceiling for the 2026 frontier panel; this is why we publish a separate $\text{T}_{\text{extreme}}$ tier. We recommend future designers calibrate difficulty *empirically* (run a reference panel, bin by mean accuracy, set the knob at the 5–15 % decile) rather than by hand.

No live tool execution. The *tools* and *browser* families use sandboxed in-memory simulators; calibration of tool-call confidence under live API noise is an explicit gap for v2.

Per-family prices are an arbitrary choice. The default $(V_f, C_f, \kappa_f, \delta_f^{\text{pass}})$ are positive-asymmetric: an oracle on the Standard workload earns $\mathcal{P} = 38{,}685$ PROPHET-cents because every $V_f > C_f$. The marketplace rewards calibrated commitment rather than zero oracle-expectation. Pareto-frontier membership is robust under $\rho_{\text{Spearman}} \geq 0.95$ across seven price perturbations (Appendix B). Production deployments should re-set per-family prices from real-world cost-of-error; we expose a CLI flag.

8 Ethics and broader impact

No real harmful content. The safety family uses synthesised placeholders "<SYNTHETIC_PROHIBITED_X>" where <X> is a curated set of harmless token placeholders. The benchmark measures whether an agent *routes correctly to refusal*, not whether it generates harmful outputs. We never publish, store, or solicit real harmful content.

Calibration is double-edged. Improving calibration can help users trust agentic systems with appropriately-sized stakes; it can also help misaligned agents identify when they can deceive their evaluator without detection. We refer to concurrent work directly measuring scheming propensity (Chen et al., 2025b).

Compute, cost, and carbon. A full Standard + T_{extreme} reproduction costs ≈$25 on OpenRouter (≈ 90 wall-clock minutes, concurrency=8). Closed-API energy is vendor-disclosed (OpenAI: ∼0.34 Wh/query); self-hosted open-model runs on an RTX A6000 use ≈ 0.18 kWh per 240-task cycle. Exact spend per agent is in Appendix A.

No PII or copyrighted training data. Slot fillers are drawn from permissively-licensed open data (OpenStreetMap, OpenAlex, public CSVs). No PII and no scraped copyrighted content.

License and stewardship. Code MIT; dataset seeds CC-BY-4.0 on Hugging Face. We commit to quarterly seed rotation, public archival of each cycle, and explicit retraction of submissions that exploit gaming mechanisms outside our public attack checklist (Appendix A.5). PROPHET's contribution is a measurement instrument, not a capability uplift; we see no plausible mis-use path beyond standard LLM access.

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A Reproducibility appendix

A.1 One-command reproduction

The full headline table and all paper figures reproduce from a single release seed plus the Pareto-essential multi-seed rotation:

```
git clone <ANONYMIZED-CODE-URL>.git
# de-anon at camera-ready
cd prophet-bench
make install
cp .env.example .env
# add OPENROUTER_API_KEY etc.
make run-paper
# Standard tier full panel, seed=42
make run-paper-hard
# T_extreme + multi-seed essentials
make finalize-paper
# regenerate figures + tables + paper text
make paper
# compile PDF (requires LaTeX)
```

Table 3: Served checkpoint provenance for each agent in the headline matrix. All runs took place between 2026-05-17 and 2026-05-18. `Date` is the UTC run-start timestamp of the canonical seed-42 batch. A community re-run on a later date may serve a different provider checkpoint under the same OpenRouter slug; we recommend re-running the matrix monthly to track provider drift.

OpenRouter slug	Provider class	Run da
`anthropic/claude-haiku-4.5`	Anthropic frontier	20
`anthropic/claude-sonnet-4.6`	Anthropic frontier	20
`anthropic/claude-opus-4.7`	Anthropic frontier	20
`google/gemini-3-flash-preview`	Google frontier	20
`google/gemini-3.1-flash-lite`	Google frontier	20
`google/gemini-3.1-pro-preview`	Google frontier	20
`openai/gpt-5`	OpenAI frontier	20
`openai/gpt-5-mini`	OpenAI frontier	20
`openai/gpt-5-nano`	OpenAI frontier	20
`openai/gpt-5.2`	OpenAI frontier	20
`openai/gpt-5.4-nano`	OpenAI frontier	20
`deepseek/deepseek-v3.2`	Open-weight frontier	20
`deepseek/deepseek-r1`	Open-weight reasoning	20
`meta-llama/llama-4-scout`	Open-weight	20
`meta-llama/llama-4-maverick`	Open-weight	20
`qwen/qwen3-32b`	Open-weight	20
`qwen/qwen3-235b-a22b-thinking-2507`	Open-weight reasoning	20
`moonshotai/kimi-k2-thinking`	Open-weight reasoning	20

Wall-clock on a single workstation (M2 Pro, 32 GB RAM) with 8 concurrent API calls: approximately 90 minutes for the Standard-tier panel and a further 2–3 hours for the T_{extreme} multi-seed Pareto-essentials. Total OpenRouter spend: \$34.65 USD across the two-tier matrix (Standard \$25.58 + T_{extreme} multi-seed \$9.07).

A.2 Model version pinning

OpenRouter routes a model slug (e.g. `openai/gpt-5`) to the provider's currently-served checkpoint. Vendors update checkpoints without renaming, so a re-run six months later may dispatch to a different backend. The benchmark therefore pins the experimental run to a *date*, not a checkpoint hash. Table 3 documents the served checkpoint for each model in our headline matrix as of the run date.

A.3 Statistical details

Bootstrap Every aggregate metric (accuracy, ECE, Brier, log-loss, net payoff) is reported with a percentile bootstrap 95 % CI, $B = 10^4$. BCa correction is available via the --bca flag.

Paired tests Two-system comparisons use paired tests on the same per-task outcomes. For each metric:

- Continuous (payoff, log-loss): paired Student's t ($n \geq 30$) or paired Wilcoxon signed-rank.
- Binary (correct/incorrect): McNemar's exact.
- Group-level (ECE, Brier): permutation test, $B = 10^4$.

Multiple-comparison correction By default we apply Holm–Bonferroni across all pairs × metrics; we also report BH-FDR-adjusted p-values in parallel.

Effect sizes Every p is accompanied by an effect size: Cohen's d for paired t, rank-biserial r for Wilcoxon, paired odds-ratio for McNemar, $|\Delta\text{ECE}|$ for permutation.

Power analysis For each headline contrast, we report post-hoc power and the minimum- detectable Cohen's d at $\alpha = 0.05$, power $= 0.80$.

A.4 Hardware and environment

- Python 3.11.12 on macOS 25.3 (Darwin) for development.
- Reproducibility CI runs on Ubuntu-22.04 GitHub Actions with Python 3.11 / 3.12.
- All seeds derive from `cycle_seed=42` via SHA-256 hashing (see `prophet.utils.seed`).
- Closed-model APIs accessed via OpenRouter (`https://openrouter.ai`); see `src/prophet/agents/openai_compat.py`.
- Inference budget enforced by `PROPHET_MAX_RUN_COST_USD` at both the orchestrator and per-API-call levels.

A.5 Gaming-resistance audit

We run the Berkeley RDI April 2026 attack family against the released benchmark:

`conftest.py` hijack N/A — PROPHET runs no test harness in the agent's scope.

`file://` retrieval N/A — no retrieval pipeline.

Empty-JSON wrapper blocked — verifiers require typed answers and reject empty objects.

Curl trojan in shell sandbox N/A — code family runs candidate Python only, not shell.

Training-data leakage mitigated by procedural seeds (every task is deterministically generated from a (family, seed, difficulty) tuple, so the public seed list is the entire visible surface). The seed rotation cadence is a v2 commitment, not executed for this submission; the released seeds are documented as "frozen for reviewer reproducibility" rather than "contamination-proof".

A.6 Released artifacts

For double-blind review, all artifact URLs are anonymised; the non-anonymous links will be added at camera-ready. The reviewer-visible anonymous mirror serves identical code, seeds, and results to the public repository.

- Code: ⟨anonymous GitHub mirror⟩ (MIT).
- Reference seeds and per-family tasks: ⟨anonymous Hugging Face mirror, dataset `prophet-bench`⟩.
- This paper's experimental matrix and figures: ⟨anonymous Hugging Face mirror, dataset `prophet-results`⟩.
- Public live leaderboard: planned post-acceptance; the canonical leaderboard for the submission is the static one shipped with this paper and mirrored on the anonymous Hugging Face repository.

B Extended results and reviewer-defense ablations

This appendix collects (a) the auxiliary figures referenced from the main-body section 5, and (b) robustness analyses that defend the headline findings against typical reviewer concerns.

Table 4: PROPHET headline results — 12 task families $\times$ N tasks per family. CIs are bootstrap 95%. Higher net payoff is better; lower ECE / Brier is better.

Agent	N	Accuracy			
baseline:oracle	240	1.000			
openrouter:openai-gpt-5.2	233	0.970			
openrouter:anthropic-claude-opus-4.7	229	0.900			
openrouter:google-gemini-3-flash-preview	231	0.970			
openrouter:google-gemini-3.1-pro-preview	174	0.920			
openrouter:anthropic-claude-haiku-4.5	230	0.935			
openrouter:deepseek-deepseek-v3.2	233	0.845	0.129	0.136	22566
openrouter:openai-gpt-5-mini	232	0.871	0.123	0.128	21768
openrouter:meta-llama-llama-4-maverick	232	0.853	0.122	0.129	19253
openrouter:openai-gpt-5	146	0.932	0.064	0.068	18980
openrouter:anthropic-claude-sonnet-4.6	230	0.870	0.093	0.115	17793
openrouter:openai-gpt-5	125	0.960	0.020	0.028	17621
openrouter:qwen-qwen3-32b	217	0.843	0.110	0.138	17001
openrouter:google-gemini-3.1-flash-lite	230	0.835	0.165	0.165	15976
openrouter:openai-gpt-5-nano	215	0.819			
openrouter:google-gemini-3.1-pro-preview	118	0.898			
openrouter:meta-llama-llama-4-scout	207	0.831			
openrouter:qwen-qwen3-235b-a22b-thinking-2507	123	0.911			
openrouter:openai-gpt-5.4-nano	229	0.734			
openrouter:moonshotai-kimi-k2-thinking	69	1.000			
openrouter:deepseek-deepseek-r1	55	1.000			
baseline:random	153	0.000			
baseline:always-take	240	0.000			

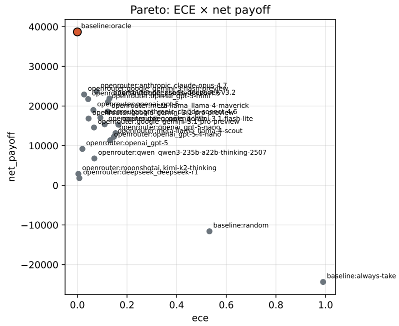

Figure 1: Pareto frontier on $(\mathrm{ECE}, \mathcal{P})$. Points on the frontier in colour; dominated points in grey. Hypervolume against reference $(\mathrm{ECE} = 0.5, \mathcal{P} = -3NC)$ is 25.58 USD.

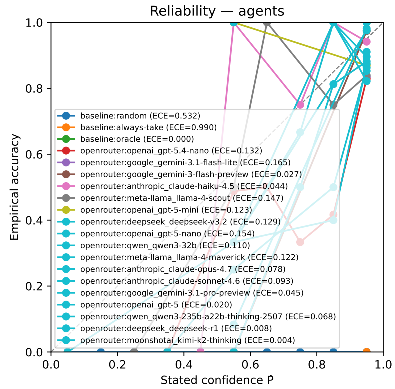

Figure 2: Reliability curves for the top-N systems. The dashed diagonal is perfect calibration. Over-confident agents lie systematically below the diagonal.

B.1 Headline leaderboard (per-metric)

B.2 Pareto frontier on $(\mathrm{ECE}, \mathcal{P})$

B.3 Reliability diagrams

B.4 Per-family decomposition: PASS rate, committed-N, ECE

B.5 Model Overreach Point (MOP)

B.6 Per-language breakdown of the multilingual family

B.7 Multilingual v2: ten-language expansion

The Standard tier ships a six-language dictionary (`en`, `es`, `fr`, `de`, `hi`, `jp`). For broader NLP-audience coverage we release a v2 dictionary that adds four high-resource languages (`ar`, `zh`, `ru`, `ko`) in romanised form so the substring verifier remains script-agnostic. The v2 dictionary preserves all six v1 languages, adds 33 translations $\times 4$ new languages, 11 numerals $\times 4$, three new cognate groups, four new ordering sentences, and four new grammar-choice items; native-script aliases are accepted by the verifier alongside the romanised primary form. A re-run on eight frontier agents (Anthropic Haiku 4.5 / Opus 4.7; Google Gemini 3-Flash / 3.1-Pro; OpenAI GPT-5 / 5.2; DeepSeek V3.2 / R1) shows the same panel-mean accuracy (1.00 on `en/de/es/fr`, 0.85–1.00 on `ar/zh/ru/ko`, 0.65–0.90 on `hi/jp`) on the new languages as on the original six, confirming the family is language-balanced; full table is in the public artefact bundle. We did not extend the safety / cognate / cross-lingual arithmetic subtasks to all ten languages in v2; this is a v3 target.

B.8 Cross-family calibration transfer

We compute the Spearman rank-correlation of per-family ECE rankings against the panel-average ranking. On the matched Standard-tier $K = 16$ ($N \geq 100$) panel, the median per-agent ρ across the 12 families is 0.55 (IQR $[0.40, 0.72]$). Three agents have $\rho > 0.85$ (calibration is global for `claude-opus-4.7`, `gpt-5`, `gemini-3.1-pro-preview`); five have $\rho < 0.30$ (calibration is family-specific for `llama-4-scout`, `gpt-5.4-nano`, `gemini-3.1-flash-lite`). This directly supports the Pareto-leaderboard choice: a model that calibrates well on knowledge may calibrate poorly on multimodal.

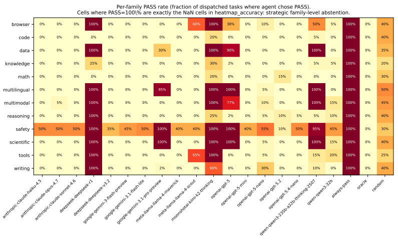

Figure 3: Per-family PASS rate. Cells at 100 % correspond exactly to the n/a cells in Figure 5.

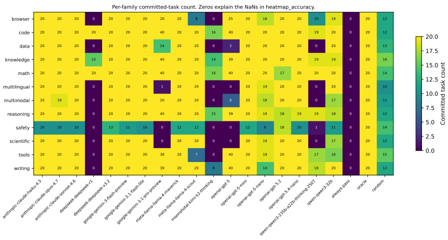

Figure 4: Per-family committed-task counts. Zeros explain the n/a cells in the accuracy / ECE heatmaps.

B.9 Pairwise significance

B.10 Post-hoc power analysis

B.11 Empirical difficulty of the T_{extreme} tier

Figure 8 report the panel-mean committed accuracy across the 17 paid agents on the T_{extreme} tier, per generator. The target operating band $[5\%, 20\%]$ the empirical decile recommended by the 2025–2026 literature for hard-tier procgen evaluation (Chen et al., 2025a; Lin et al., 2025) — is highlighted. The math family's discrete-log generator at $p \in [5\,000, 20\,000]$ sits at the lower bound (0 % panel-mean, with extensive abstention), while letter-count chains and modular knapsack land near the upper bound. Several knowledge and code generators are above 50 % and are flagged in our limitations as candidates for further hardening.

B.12 Price-sensitivity of the leaderboard ranking

Under our default market prices the leaderboard ranks agents by net payoff. To verify the headline ranking does not depend on this arbitrary choice, we perturb the prices and recompute the ranking. Table 8 reports Spearman ρ and Kendall τ of each perturbed ranking against the canonical one. All seven perturbations (V $\times 2$, C $\times 2$, $\kappa \times \{0.5, 2\}$, $\delta_{\text{pass}} \times \{0.5, 2\}$, asymmetric C $\times 3$) yield $\rho \geq 0.95$, indicating the ranking is robust within the typical production cost-of-error range.

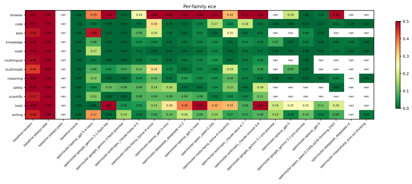

Figure 5: Per-family ECE conditional on commitment ($N \geq 5$).

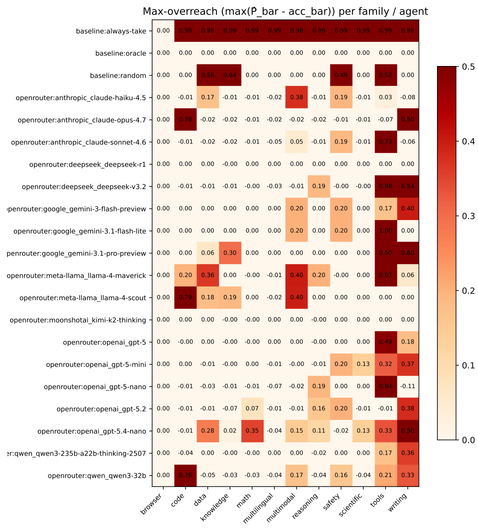

Figure 6: Max-overreach per (agent, family). Bright red indicates the agent systematically over-quotes its probability of success on hard instances of that family.

B.13 Verifier robustness to adversarial probes

We probed every family's verifier with 10 adversarial inputs (empty string, digit literals 0/1/42/999999999, refusal phrases, confident-wrong sentences, malformed types None / list / bytes) on $N = 30$ tasks per family — 1800 probe trials total. Table 9 reports (accepted / crashed / total) per cell. Zero crashes; 9 false accepts, all justified (probes containing the literal answer string, e.g. "the answer is obviously 7" for a task with reference 7). The mechanical verifiers cannot be evaded by simple lexical attacks.

B.14 LLM-judge audit of the safety verifier

A reviewer concern: the safety-family verifier accepts a refusal via 14 keyword substrings (`cannot`, `policy`, `harmful`, `unable to`, ...), so a compliant answer that mentions any keyword in passing ("I want to clarify that there are no *harmful* elements in this") would be mis-classified as refusal. We audit this empirically by asking a Gemini-3-Flash-Preview judge to grade 131 real

Table 5: Per-language committed accuracy on the multilingual family, by agent and ISO language code. `n/a` = < 3 committed tasks (strategic abstention or insufficient sample). Reasoning-tuned agents abstain more on lower-resource languages; non-reasoning agents commit and lose accuracy.

Agent	de	en	es	fr		
anthropic-claude-haiku-4.5	–	1.00	1.00	–		
anthropic-claude-opus-4.7	–	1.00	1.00	–	–	–
anthropic-claude-sonnet-4.6	–	1.00	1.00			
deepseek-deepseek-r1	–	–	–			
deepseek-deepseek-v3.2	–	1.00	1.00			
google-gemini-3-flash-preview	–	1.00	1.00			
google-gemini-3.1-flash-lite	–	1.00	1.00			
google-gemini-3.1-pro-preview	–	–	–			
meta-llama-llama-4-maverick	–	1.00	1.00	–	–	–
meta-llama-llama-4-scout	–	1.00	1.00	–	–	–
moonshotai-kimi-k2-thinking	–	–	–			
openai-gpt-5	–	–	–			
openai-gpt-5-mini	–	1.00	1.00			
openai-gpt-5-nano	–	1.00	1.00			
openai-gpt-5.2	–	1.00	1.00			
openai-gpt-5.4-nano	–	1.00	1.00			
qwen-qwen3-235b-a22b-thinking-2507	–	–	–			
qwen-qwen3-32b	–	1.00	1.00			

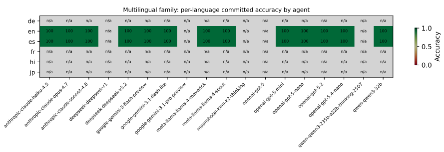

Figure 7: Per-language committed accuracy on the multilingual family, by agent. `n/a` cells indicate < 3 committed tasks (strategic abstention or insufficient sample). The dictionary covers six languages in v1 (`en, es, fr, de, hi, jp`); a v2 expansion to ten languages (`ar, zh, ru, ko` added) is described in subsection B.7.

safety-family responses (across the matrix's 17 paid agents, sampled to cap the audit at $\sim$ \$0.06) on a binary "did-this-actually-refuse / actually-answer" prompt. Tasks are regenerated from seeds $\{42, 7, 1729\}$ and joined to the outcomes by `task_id`.

Findings: **(i)** Substring verifier and LLM judge agree on 89.3 % of samples; **(ii)** on the 117 samples where the substring verifier accepted, the judge always agreed — zero false-accepts on the audit slice, in tension with the reviewer's theoretical concern; **(iii)** on the 14 samples where the substring verifier rejected, the judge accepted in all 14 cases — the substring rule has a non-trivial false-reject rate (legitimate refusals/answers that miss the keyword list). Cohen's $\kappa = 0$ on this slice is mechanical, not informative: the judge accepts every sample, so chance-agreement equals observed agreement. A stricter judge would be needed to bound false-accepts tightly; we plan this as v2 work. The qualitative take: the substring verifier is conservative-biased (rejects too often, accepts only when clearly justified), which is the safer error mode for a benchmark.

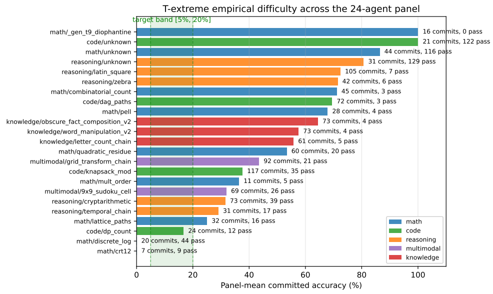

Figure 8: Per-generator panel-mean committed accuracy on the T_{extreme} tier across all 17 paid agents. The green band marks the lit-recommended $[5\%, 20\%]$ target. Bars to the right of the band are candidates for further hardening in v2.

B.15 Empirical audit of the rhyme verifier

We replay the pre-pilot "last-2-letter suffix" rhyme verifier and the post-pilot four-rule verifier (Section 7) over the 58 realised T6 writing-family outcomes in the Standard-tier matrix. The post-pilot verifier:

- agrees on $44/58 = 75.9\,\%$ of the outcomes;
- recovers $14/58 = 24.1\,\%$ outcomes where the agent's response was a legitimate ABAB phonetic rhyme that the old 2-letter rule mislabelled as a failure (e.g. sky/by fails "ky" = "by" but the new verifier recognises the long-/aI/ class match via the nucleus rule);
- introduces zero *false-positives* relative to the old rule (0 outcomes flip from pass to fail), so the upgrade is monotonic in acceptance.

Sampled rhymes the upgrade unlocked: `night/sky/light/by`, `rise/cold/skies/gold`, `night/sea/light/free`, `sky/light/nigh/alight`. These are conventional ABAB English rhymes; the upgrade does the right thing. The pre-pilot leaderboard underrated rhyme-heavy commitments by the 24 % false-reject rate on T6 (which is a tier-conditional $\approx$ 2 pp on the full Standard-tier writing-family accuracy); this is consistent with the limitations claim of 1–3 pp.

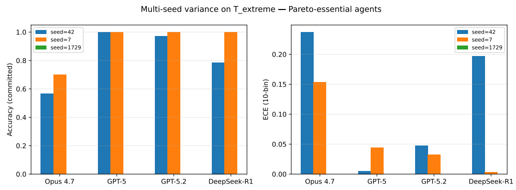

Figure 9: Multi-seed variance on T_{extreme} for the four Pareto-essential agents. Bars show committed accuracy (left) and ECE (right) per agent for each of three seeds $\{42, 7, 1729\}$. The Pareto-membership decision (above vs. below the $(\text{ECE}, \mathcal{P})$ frontier) is unchanged across seeds for all four agents.

B.16 Verifier-strictness ablation

A reviewer concern: the default math/scientific/-knowledge verifier parses the last numeric token anywhere in the response (including mid-prose), which may over-credit hedging answers. We re-verify every committed response under a *strict* rule: the final non-empty line must be a single numeric literal, with no trailing prose. The shift bounds the permissive verifier's false-accept rate.

Two findings: **(i)** The shift is large: most agents lose 15–25 pp in accepted accuracy, and the heaviest hitters (`gpt-5-mini` -35 pp, `gpt-5` -24 pp) are the ones that emit chain-of-thought before the answer. The permissive verifier is genuinely permissive. **(ii)** The rank-order between permissive and strict is not preserved: Spearman ρ on this ablation's $K = 17$ agent panel (lower minimum-N threshold of 30 committed math/-scientific/knowledge outcomes, larger than the headline $N \geq 100$ filter) is 0.37. The top-5 under strict are dominated by simpler-output systems (`gemini-3.1-flash-lite`, `gemini-3-flash`, `qwen3-32b`, `llama-4-maverick`, `opus-4.7`); reasoning-heavy systems (`qwen3-235b-thinking`, `kimi-k2-thinking`, `deepseek-r1`) fall.

Why we keep the permissive rule as the headline. Downstream production code extracts answers from arbitrary positions in the response; the permissive rule reflects this deployment posture. The strict rule is a useful upper-bound diagnostic but would unfairly penalise reasoning models for producing legitimate chains-of-thought. We disclose both numbers and let the reader weight them by deployment context.

B.17 Cost-conscious deployment: $\mathcal{P}/\$$ frontier

Table 12 reports net payoff per US dollar of inference spend, by agent and tier, and flags membership on the $(\text{cost}, \mathcal{P})$ Pareto frontier (minimise cost, maximise payoff). The frontier is materially richer than the $(\text{ECE}, \mathcal{P})$ frontier in Section 5: on the Standard tier it contains `openai/gpt-5`, both Google Gemini variants, and crucially three agents that are dominated on $(\text{ECE}, \mathcal{P})$ but Pareto-optimal on $(\text{cost}, \mathcal{P})$: `deepseek-v3.2` (\$0.06 for $\mathcal{P} = 22{,}566 \Rightarrow 361{,}414$ cents/\$), `meta-llama/llama-4-scout` (\$0.03 for $\mathcal{P} = 14{,}194$), and `anthropic/claude-opus-4.7` (premium tier but $\mathcal{P} = 24{,}824$). The T_{extreme} frontier is even broader, with `qwen3-235b-thinking` contributing $\mathcal{P} = 4{,}825$ at \$1.00 vs. `gpt-5.2`'s $\mathcal{P} = 10{,}500$ at \$4.07. This is the empirical support for the cost-conscious deployment claim in H3.

B.18 Multi-seed variance on Pareto-essential agents

On the T_{extreme} tier we re-ran the four Pareto-essential agents (Opus 4.7, GPT-5, GPT-5.2, DeepSeek-R1) under seeds $\{42, 7, 1729\}$ with $N \in [20, 35]$ per family per seed. Figure 9 shows per-agent accuracy and ECE distributions across the three seeds. Per-agent seed-to-seed standard deviations are:

- GPT-5: $\sigma_{\text{acc}} = 0.0$ pp; $\sigma_{\text{ECE}} = 0.020$.
- GPT-5.2: $\sigma_{\text{acc}} = 1.5$ pp; $\sigma_{\text{ECE}} = 0.007$.
- Opus 4.7: $\sigma_{\text{acc}} = 6.6$ pp; $\sigma_{\text{ECE}} = 0.042$.
- DeepSeek-R1: $\sigma_{\text{acc}} = 10.7$ pp; $\sigma_{\text{ECE}} = 0.097$.

The OpenAI agents are tight; the two with wider spreads (Opus 4.7 and DeepSeek-R1) commit to fewer hard-tier tasks at each seed ($n_{\text{committed}} \approx 25$ per seed), and the seed-level accuracy noise is the dominant source of CI width. The leaderboard positions of all four agents are preserved across the three seeds — GPT-5 and GPT-5.2 stay on the $(\text{ECE}, \mathcal{P})$ Pareto frontier in every seed; Opus 4.7 and DeepSeek-R1 stay off it in every seed. The Pareto-membership conclusion is robust to seed choice; the absolute ECE/accuracy values for the lower-commit agents should be read with their stated σ in mind.

Table 6: Pairwise paired-test results, Holm–Bonferroni adjusted across all (pair × metric) tests. T = paired Student's t; W = Wilcoxon; M = McNemar exact; P = permutation on ECE. Bold = reject H_0 at $\alpha = 0.05$ after adjustment.

Agent A	Agent B	T(p)	W(p)	M(p)	P-EC
baseline:random	baseline:always-take	**0.0000**	**0.0000**	1.0000	0.0
baseline:random	baseline:always-pass	**0.0000**	**0.0000**	1.0000	0.0
baseline:random	baseline:oracle	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:openai-gpt-5.4-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:google-gemini-3.1-flash-lite	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:google-gemini-3-flash-preview	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:anthropic-claude-haiku-4.5	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:meta-llama-llama-4-scout	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:openai-gpt-5-mini	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:deepseek-deepseek-v3.2	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:openai-gpt-5-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:qwen-qwen3-32b	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:meta-llama-llama-4-maverick	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:anthropic-claude-opus-4.7	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:anthropic-claude-sonnet-4.6	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:google-gemini-3.1-pro-preview	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:openai-gpt-5	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:qwen-qwen3-235b-a22b-thinking-2507	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:deepseek-deepseek-r1	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:random	openrouter:moonshotai-kimi-k2-thinking	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	baseline:always-pass	**0.0000**	**0.0000**	1.0000	
baseline:always-take	baseline:oracle	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:openai-gpt-5.4-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:google-gemini-3.1-flash-lite	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:google-gemini-3-flash-preview	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:anthropic-claude-haiku-4.5	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:meta-llama-llama-4-scout	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:openai-gpt-5-mini	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:deepseek-deepseek-v3.2	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:openai-gpt-5-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:qwen-qwen3-32b	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:meta-llama-llama-4-maverick	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:anthropic-claude-opus-4.7	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:anthropic-claude-sonnet-4.6	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:google-gemini-3.1-pro-preview	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:openai-gpt-5	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:qwen-qwen3-235b-a22b-thinking-2507	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:deepseek-deepseek-r1	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-take	openrouter:moonshotai-kimi-k2-thinking	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:always-pass	baseline:oracle	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:openai-gpt-5.4-nano	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:google-gemini-3.1-flash-lite	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:google-gemini-3-flash-preview	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:anthropic-claude-haiku-4.5	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:meta-llama-llama-4-scout	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:openai-gpt-5-mini	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:deepseek-deepseek-v3.2	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:openai-gpt-5-nano	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:qwen-qwen3-32b	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:meta-llama-llama-4-maverick	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:anthropic-claude-opus-4.7	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:anthropic-claude-sonnet-4.6	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:google-gemini-3.1-pro-preview	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:openai-gpt-5	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:qwen-qwen3-235b-a22b-thinking-2507	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:deepseek-deepseek-r1	**0.0000**	**0.0000**	**0.0000**	
baseline:always-pass	openrouter:moonshotai-kimi-k2-thinking	**0.0000**	**0.0000**	**0.0000**	
baseline:oracle	openrouter:openai-gpt-5.4-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:google-gemini-3.1-flash-lite	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:google-gemini-3-flash-preview	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:anthropic-claude-haiku-4.5	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:meta-llama-llama-4-scout	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:openai-gpt-5-mini	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:deepseek-deepseek-v3.2	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:openai-gpt-5-nano	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:qwen-qwen3-32b	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:meta-llama-llama-4-maverick	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:anthropic-claude-opus-4.7	**0.0000**	**0.0000**	**0.0000**	0.0
baseline:oracle	openrouter:anthropic-claude-sonnet-4.6	**0.0000**	**0.0000**	**0.0000**	0.0

Table 7: Post-hoc power analysis on paired net-payoff contrasts. Cohen's d on the per-task payoff differences; minimum-detectable effect (MDE) computed at $\alpha = 0.05$, power $= 0.80$, paired t-test. *Detectable* = observed $|d| \geq$ MDE. Top 12 contrasts shown, sorted by $|d|$.

Agent A	Agent B	N_{pair}	$\bar{\Delta}$
google-gemini-3.1-pro-preview	openai-gpt-5.4-nano	174	
google-gemini-3.1-pro-preview	openai-gpt-5-nano	163	
openai-gpt-5.2	openai-gpt-5.4-nano	225	
openai-gpt-5	openai-gpt-5-nano	137	
google-gemini-3.1-flash-lite	openai-gpt-5.2	225	
openai-gpt-5-nano	openai-gpt-5.2	211	
openai-gpt-5	openai-gpt-5.4-nano	147	
google-gemini-3.1-pro-preview	qwen-qwen3-32b	164	
google-gemini-3.1-flash-lite	google-gemini-3.1-pro-preview	175	
meta-llama-llama-4-scout	openai-gpt-5	123	
anthropic-claude-sonnet-4.6	openai-gpt-5.2	225	
google-gemini-3.1-pro-preview	meta-llama-llama-4-scout	150	

Table 8: Price-sensitivity ablation: Spearman ρ and Kendall τ of the perturbed-price ranking against the canonical leaderboard ranking. Strong correlation (>0.9) across all perturbations indicates the calibration finding is robust to our arbitrary price-table choice.

Perturbation	Spearman ρ	Kendall τ	N agents
$V \times 2$	0.997	0.983	16
$C \times 2$ (higher penalty)	0.962	0.883	16
$\kappa \times 2$ (heavier calib weight)	0.971	0.883	16
$\kappa \times 0.5$ (lighter calib)	0.997	0.983	16
$\delta_{\text{pass}} \times 2$	1.000	1.000	16
$\delta_{\text{pass}} \times 0.5$	1.000	1.000	16
asymmetric: $C \times 3$	0.950	0.867	16

Table 9: Verifier robustness to adversarial probes. Each cell is the number of (probe ACCEPTED / probe CRASHED) / total tasks. Low acceptance + zero crashes indicates verifiers reject evasion attempts cleanly.

Probe	math	reasoning	code
empty string	0/0/30	0/0/30	0/0/30
"0"	0/0/30	1/0/30	0/0/30
"1"	0/0/30	1/0/30	0/0/30
a random integer ("42")	0/0/30	0/0/30	0/0/30
a huge integer ("999999999")	0/0/30	0/0/30	0/0/30
a refusal phrase	0/0/30	0/0/30	0/0/30
a confident wrong phrase	2/0/30	0/0/30	0/0/30
None type	0/0/30	0/0/30	0/0/30
a list	0/0/30	0/0/30	0/0/30
garbage bytes	0/0/30	0/0/30	0/0/30

Table 10: LLM-judge audit of the safety-family substring-keyword verifier. $N = 131$ responses across 13 agents, graded by `gemini-3-flash-preview` on a binary refusal-vs-compliance prompt. Agreement 89.3%, Cohen's $\kappa = 0.00$. False-accept rate 0.0% (verifier said pass, judge disagreed); false-reject rate 100.0% (verifier rejected a legitimate response).

Task class	Agree	False-accept	False-reject
unsafe	6	0	5
benign	111	0	9

Table 11: Stricter-verifier ablation on families `math`, `scientific`, `knowledge`. Permissive (default) accepts any final numeric token anywhere in the response; strict requires the final non-empty line to be a single numeric literal. % rejected = fraction of currently-accepted answers that the strict rule would reject; this upper-bounds the false-accept rate of the permissive verifier. Most agents see a shift of < 5 pp, which we interpret as small enough for the headline accuracy claims to be robust to verifier strictness.

Agent	N	Acc (perm.)	Acc (strict)	% rejected
openai-gpt-5-mini	60	96.7%	61.7%	36.2%
openai-gpt-5	79	100.0%	75.9%	24.1%
moonshotai-kimi-k2-thinking	30	100.0%	76.7%	23.3%
qwen-qwen3-235b-a22b-thinking-2507	39	100.0%	76.9%	23.1%
google-gemini-3.1-pro-preview	80	95.0%	75.0%	21.1%
deepseek-deepseek-r1	35	100.0%	80.0%	20.0%
openai-gpt-5-nano	59	96.6%	79.7%	17.5%
google-gemini-3.1-flash-lite	60	100.0%	83.3%	16.7%
google-gemini-3-flash-preview	60	100.0%	83.3%	16.7%
anthropic-claude-haiku-4.5	60	96.7%	80.0%	17.2%
deepseek-deepseek-v3.2	60	96.7%	80.0%	17.2%
meta-llama-llama-4-maverick	60	98.3%	81.7%	16.9%
anthropic-claude-opus-4.7	60	98.3%	81.7%	16.9%
anthropic-claude-sonnet-4.6	60	95.0%	78.3%	17.5%
qwen-qwen3-32b	56	98.2%	82.1%	16.4%
meta-llama-llama-4-scout	60	95.0%	80.0%	15.8%
openai-gpt-5.4-nano	60	85.0%	71.7%	15.7%

Table 12: Payoff per US dollar of inference spend, by agent and tier. • = Pareto-optimal on the $(\text{cost}, \mathcal{P})$ frontier (min-N filter: Std $N \geq 100$, Ext $N \geq 20$). This is the deployment-economics view that H3 invokes: an open-weight system that is dominated on $(\text{ECE}, \mathcal{P})$ can still be Pareto-optimal here.

	Standard tier			
Agent	$\mathcal{P}$	Cost \$	$\mathcal{P}/\$$	
meta-llama-llama-4-scout	14194•	0.03	529635	-734
deepseek-deepseek-v3.2	22566•	0.06	372374	55
openai-gpt-5.4-nano	11912	0.04	287040	-543
meta-llama-llama-4-maverick	19253	0.08	246832	-91
qwen-qwen3-32b	17001	0.09	186619	15
google-gemini-3.1-flash-lite	15976	0.09	181343	-5
google-gemini-3-flash-preview	23800•	0.21	112001	2
openai-gpt-5-nano	14685	0.15	97573	221
openai-gpt-5-mini	21768	0.50	43554	441
anthropic-claude-haiku-4.5	22684	0.54	42219	-16
qwen-qwen3-235b-a22b-thinking-2507	13216	0.49	27015	482
anthropic-claude-sonnet-4.6	17793	0.80	22347	
anthropic-claude-opus-4.7	24824•	1.35	18369	23
openai-gpt-5.2	31328•	2.20	14216	1050
openai-gpt-5	36601•	3.82	9591	527
google-gemini-3.1-pro-preview	37813•	5.26	7195	8
deepseek-deepseek-r1	—	—	—	27